UPDATED PROJECT BRIEF | POSTER SOURCE

Civic technology | NLP-assisted triage

CivicSense

Public Grievance Intelligence Platform

Project overview. CivicSense is a web platform for residents to report local civic problems, follow their progress, and coordinate community support. Staff receive a structured review queue with advisory NLP signals to help prioritize human review.

PROBLEM

Local service issues can be difficult to report and track consistently. Residents need a clear way to submit evidence and see updates, while reviewers need useful context to triage incoming reports.

SOLUTION & KEY FEATURES

- Account registration and sign-in; optional photo and map location
- Personal report history, status timelines, and notifications
- Community map, search, filters, updates, and report support
- Admin triage: category, department, priority, response target, approval, decline reason, and resolution

HOW IT WORKS

Residents submit a report, optionally attaching a photo and shared location. The service analyzes text locally and saves advisory signals. Admins review the report and NLP signals, take a human decision, and residents can follow status changes.

Resident report > **Local NLP assistance** > **Admin human review** > **Status & community follow-up**

NLP CONTRIBUTION

Deterministic, rules-based text analysis runs locally. It detects supported writing systems (English, Hindi, Gujarati), suggests a category, highlights urgency, creates an extractive summary and place hints, and finds similar public reports.

Spam screening: checks promotional/scam wording, links, irrelevant phrases, repetitive wording, and long repeated characters. The admin queue shows a clear possible-spam result and reasons; existing reports are screened during database startup migration.

TECHNOLOGY

Frontend: React 18, Vite
Backend: Python, FastAPI
Database: SQLite
Maps: Leaflet / React Leaflet with OpenStreetMap tiles
NLP: in-house local Python rules; no external AI service or trained model.

VERIFIED IMPLEMENTATION

End-to-end reporting, review, and community workflows are implemented. The current NLP suite has **9 passing unit tests**; the frontend production build passes. These are software check - not model accuracy or a field study.

EVALUATION & LIMITATIONS

No validated classification-accuracy score, response-time benchmark, deployment/user count, or evaluated complaint dataset is currently available. Do not invent metrics. NLP signals are heuristic suggestions, may be wrong, and never automatically reject reports or decide priority.

POSTER DESIGN DIRECTION

Use a clean academic-poster layout with a civic/community feel: deep forest green, warm off-white, and restrained leaf-green accents. Make the resident > NLP > reviewer - follow-up workflow the central visual. Include a small mock report card showing a possible-spam flag with reasons, a map/report illustration, and clearly label NLP as rules-based decision suppor | ot a trained AI classifier. Keep text concise and avoid fabricated results.

Presenters: [Add names] **Guide:** [Add name]

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